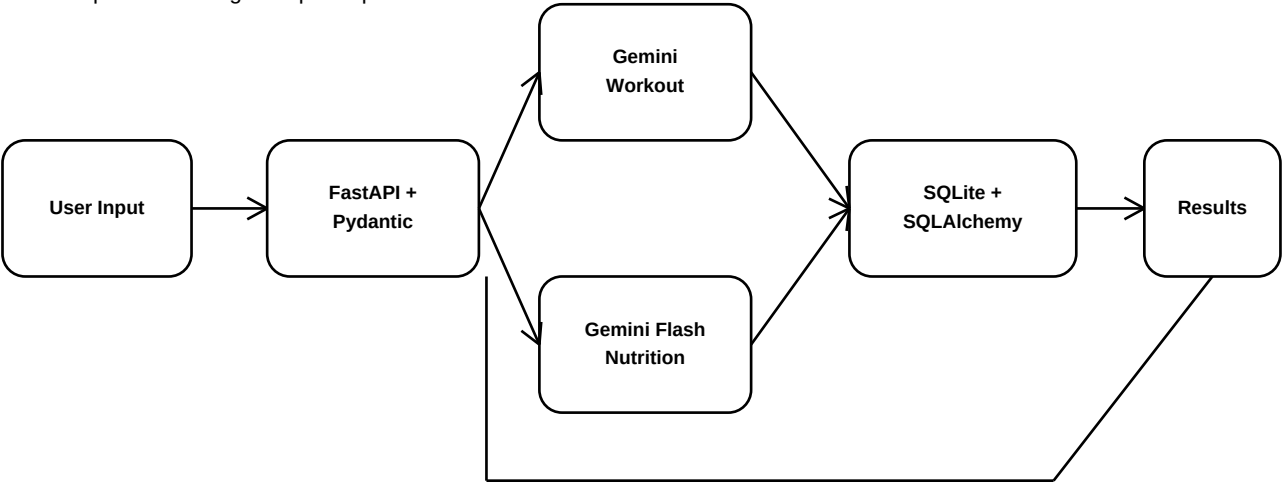


Solution Architecture

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	5 Marks

Solution Architecture

FitBuddy follows a layered request path from the browser to FastAPI, the Gemini generation layer and SQLite storage. Feedback loops back through the plan-update AI module.



Feedback is submitted from Results and sent back to the AI update module.

Layer Responsibilities

Layer	Components	Responsibility
Presentation	HTML, CSS, Jinja2	Collect inputs and display generated plans, tips and navigation.
Application	FastAPI routes, Pydantic schemas	Validate forms, orchestrate AI calls, manage responses and errors.
AI	Gemini generator, Gemini Flash generator, updated plan	Generate workout plan, nutrition/recovery tip and revised plan.
Persistence	SQLite + SQLAlchemy	Store user profiles, original plans, feedback and updated plans.
Runtime	Uvicorn	Serve the FastAPI application locally.